

Support slice + mask

Query slice

**Shared lightweight ViT backbone
(MobileViT-S / ViT-Tiny) → dense tokens**

SSP module
boundary band = dilation-erosion
→ mini-batch k -means
→ hard boundary prototypes

Global FG / BG
prototypes
(masked pooling)

Cosine-similarity matching
channels: [global-FG, max-hard, BG]

1×1 conv + sigmoid

Predicted query mask

Objective: $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{boundary}}$ (Sobel GT boundary)